# Dinesh Kumar Yadav

Data Science | Machine Learning | AI Engineer
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LinkedIn | GitHub

## Professional Summary

Computer Science undergraduate focused on **Machine Learning, Deep Learning and Generative AI**, with research experience in **Medical Image Segmentation and Quantum Machine Learning** at Symbiosis International University. Experienced in ML preprocessing, model training, evaluation, RAG systems and deployment using **Python and FastAPI**. Twice selected as **Team Lead** for institute-level Smart India Hackathon.

## Technical Skills

**Programming:** Python, C, C++, SQL
**Machine Learning:** Scikit-learn, Pandas, NumPy, Classification, Regression, Clustering, Feature Engineering, Cross-Validation
**Deep Learning:** TensorFlow, Keras, PyTorch, CNN, U-Net, Transfer Learning, Image Segmentation, Data Augmentation, OpenCV
**Generative AI:** LLMs, RAG, Prompt Engineering, Vector Embeddings, Semantic Search, Conversational AI
**Quantum ML:** Quantum Machine Learning, Variational Quantum Circuits, Hybrid Quantum-Classical Models
**Deployment & Tools:** FastAPI, REST APIs, Docker, Git/GitHub, Jupyter, Google Colab, Postman
**Evaluation:** Accuracy, Precision, Recall, F1-Score, ROC-AUC, Confusion Matrix, Dice, IoU

## Research Experience

**Research Intern – Quantum Machine Learning** Aug 2026 – Present
*Symbiosis International University (SIU)* *Pune, India*

- Implementing **variational quantum circuits and hybrid quantum-classical models** for classification tasks and benchmarking them against classical baselines.
- Studying quantum data encoding techniques for mapping classical datasets into quantum feature spaces.

**Research Intern – Medical Image Segmentation** June 2026 – July 2026
*Symbiosis International University (SIU)* *Pune, India*

- Developed a **U-Net model** for automated Meibomian gland segmentation from infrared meibography images.
- Built preprocessing and augmentation pipelines and evaluated segmentation using **Dice Coefficient and IoU**.

## AI & Machine Learning Projects

**MCAET College Information Assistant** | *Generative AI, RAG, LLM, Python* | Live Website | GitHub 2026

- Built a **RAG-based conversational assistant** for college courses, admissions, departments, facilities and campus information.
- Implemented document chunking, embeddings and semantic search to retrieve relevant institutional context and exposed the system through a REST API.

**Facial Emotion Recognition** | *CNN, TensorFlow/Keras, OpenCV* | GitHub Jan 2026

- Trained a CNN to classify **6 facial emotions** from 4,500+ grayscale images, achieving **89% accuracy and 0.87 weighted F1-score**.
- Applied data augmentation and dropout regularization and analyzed confusion-matrix based misclassification patterns.

**Cattle Breed Classification** | *CNN, Transfer Learning, Computer Vision* | GitHub 2025

- Developed a deep learning classifier for multiple Indian cattle breeds, achieving **80% accuracy** on the evaluation dataset.
- Applied preprocessing and augmentation to improve generalization on an imbalanced image dataset.

## Achievements & Leadership

- **Smart India Hackathon (SIH):** Cleared institute-level internal round twice (2025, 2026); served as **Team Lead**.
- Selected for two consecutive research internships at **Symbiosis International University** in applied AI domains.

## Education

**Acharya Narendra Deva University of Agriculture & Technology (ANDUAT)** Ayodhya, UP
*B.Tech, Computer Science and Engineering* | ***CGPA: 7.5/10*** *Aug 2024 – May 2028*

## Certifications

- Python for Data Science – Simplilearn
- AI Predict (Python) Workshop – SkillManthan Technologies Pvt. Ltd.
- Participant, AlgoUtsav 2026 Hackathon